

University of Pisa
Department of Mathematics
Master's Degree in Mathematics

Algebraic Number Theory 3

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1. p-adic number fields

Definition 1.0.1 (Brauer group)

Let k be a field and k_s be one of its separable closures. The *Brauer group* of k is

$$\mathrm{Br}(k) := H^2(k, k_s)$$

If $K|k$ is a separable subextension of $k_s|k$ we can define the *relative Brauer group* of the extension as

$$\mathrm{Br}(K|k) := \ker\left(\mathrm{Br}(k) \xrightarrow{\mathrm{Res}} \mathrm{Br}(K)\right)$$

The elements of $\mathrm{Br}(k)$ correspond to the finite dimensional division algebras D with center k. In this correspondence, the subgroup $\mathrm{Br}(K|k)$ gives those D for which $D \otimes_k K \simeq M_{n\times n}(K)$, where $n^2 = \dim_k D$.

Lemma 1.0.1

1. If $K|k$ is a Galois extension with group G then it holds $\mathrm{Br}(K|k) \simeq H^2(G, K^\times)$;
2. The Brauer group $\mathrm{Br}(k)$ is the union of all the subgroups $\mathrm{Br}(K|k)$ for finite Galois extensions $K|k$;
3. If G is finite and cyclic it holds $\mathrm{Br}(K|k) \simeq \frac{k^\times}{N_{K|k}(K^\times)}$.

Here $N_{K|k} : K^\times \to k^\times$ denotes the norm of the field extension. When $K|k$ is finite and Galois it has value

$$N_{K|k}(a) = \prod_{\sigma\in \mathrm{Gal}(K|k)} \sigma a$$

for every $a \in K^\times$.

Proof:

1. From Hilbert's theorem 90 we have $H^1(G, K^\times) = 0$. As seen in # this makes the following inflation-restriction sequence exact:

   $$0 \longrightarrow H^2(G, K^\times) \xrightarrow{\mathrm{Inf}} H^2(k, k_s) \xrightarrow{\mathrm{Res}} H^2(K, k_s),$$

   and by definition $\mathrm{Br}(K|k) = \ker \mathrm{Res} = H^2(G, K^\times)$.
2. From the previous point we deduce

   $$\mathrm{Br}(k) = \varinjlim H^2(G, K^\times) \overset{(1)}{=} \varinjlim \mathrm{Br}(K|k) = \bigcup \mathrm{Br}(K|k).$$
3. As seen in # when the Galois group G is cyclic we have

   $$\mathrm{Br}(K|k) \overset{(1)}{=} H^2(G, K^\times) = \frac{(K^\times)^G}{N_{K|k}(K^\times)} = \frac{k^\times}{N_{K|k}(K^\times)}.$$

□

Example 1.0.1

0. When the field k is complete we have $\mathrm{Br}(k) = 0$ since the Galois action on it is trivial.
1. Let's compute the Brauer group of the real numbers:

$$\mathrm{Br}(\mathbb{R}) = \mathrm{Br}(\mathbb{C}|\mathbb{R}) = H^2(\mathrm{Gal}(\mathbb{C}|\mathbb{R}), \mathbb{C}^\times) = \frac{\mathbb{R}^\times}{N_{\mathbb{C}|\mathbb{R}}(\mathbb{C}^\times)} = \frac{\mathbb{R}^\times}{\mathbb{R}^\times_{>0}} \simeq \mathbb{Z}/2\mathbb{Z}.$$

2. When $k = \mathbb{F}_q$ is a finite field the Brauer group vanishes: $\mathrm{Br}(\mathbb{F}_q) = 0$. By the lemma 2) it is enough to show that $\mathrm{Br}(\mathbb{F}_{q^r}|\mathbb{F}_q) = 0$ for every $r > 0$. Recall the following facts:
   - The group $\mathrm{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_q)$ is cyclic, generated by the frobenius $a \mapsto a^q$. In particular the norm has value
   $$N(a) = a^{\frac{q^r-1}{q-1}};$$
   - The multiplicative group $\mathbb{F}_{q^r}^\times$ is cyclic.

   Let ω be a generator of $\mathbb{F}_{q^r}^\times$, its norm $N(\omega)$ by the above calculation has order $q-1$ and therefore generates $\mathbb{F}_q$. This shows N is surjective and by lemma 3) we deduce $\mathrm{Br}(\mathbb{F}_{q^r}|\mathbb{F}_q) = 0$.

1.1. The path to $K|\mathbb{Q}_p$

The main goal of this section is proving that if $K|\mathbb{Q}_p$ is a p-adic number field then its Brauer group is $\mathrm{Br}(K) = \mathbb{Q}/\mathbb{Z}$.

Definition 1.1.2